

Stephen Kamau

Data Engineer | **ML Deployment** | **Cloud Data Platforms** | **ETL and Pipeline Architecture**

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PROFESSIONAL PROFILE

Data Engineer with over three years of hands-on experience designing, building, and maintaining production-grade data platforms across financial services, healthcare, and logistics. Skilled in architecting end-to-end pipelines from raw data ingestion and ETL transformation using Airflow and dbt, through warehouse modelling on Snowflake, to deploying machine learning models on AWS in regulated, high-stakes environments. Works effectively within cross-functional engineering and analytics teams, translating complex business requirements into scalable, maintainable data infrastructure. Brings both technical rigour and a creativity-driven mindset, and is energised by the opportunity to power data-driven decisions inside one of Kenya's most trusted financial institutions.

TECHNICAL SKILLS

SQL and Databases PostgreSQL, MySQL, Snowflake, advanced query authoring, indexing, query optimisation

Data Warehousing Snowflake, HDFS, S3-based data lakes, schema design, partitioning, data modelling

ML Deployment AWS SageMaker, Lambda, EKS — model serving, monitoring, resource configuration

BI and Analytics Metabase, Looker, Streamlit, custom dashboards, KPI framework design

Programming Python (primary), SQL, Java, C++

ETL and ELT Tools Apache Airflow, dbt, Fivetran, Luigi, custom Python ETL pipelines

Big Data Apache Spark, HDFS, distributed batch and streaming pipeline processing

Cloud and Infra AWS (EC2, S3, Lambda, SageMaker, EKS), Docker, Kubernetes, GitHub Actions

Monitoring CloudWatch, Prometheus, Grafana, SLA alerting, pipeline health and model drift

ML and Data Science Scikit-learn, XGBoost, PyTorch, BERT, time-series forecasting, statistical modelling

PROFESSIONAL EXPERIENCE

Prospect33 *Data Scientist and ML Engineer* Jan 2024 to Present

Prospect33 delivers AI-powered data intelligence solutions to top-tier U.S. financial institutions, automating extraction and analytics on large volumes of legal and financial documents.

- **Architected end-to-end data pipelines** using Airflow, dbt, and Snowflake for a regulated U.S. banking client, covering ingestion, transformation, quality validation, and delivery to downstream analytics systems at scale.
- Designed and maintained **ETL workflows** that ingested, cleaned, and structured unstructured financial document data, with schema validation and quality checks to ensure reliability in a compliance-sensitive environment.
- **Deployed and monitored ML models** on AWS SageMaker, Lambda, and EKS, configuring compute resources, managing model versions, and setting up SLA-based alerting via CloudWatch to maintain production uptime targets.
- Built and deployed **OCR pipelines using Tesseract and AWS Lambda**, automating extraction of structured data from image-heavy PDFs and reducing manual data entry effort by **80%**, with clean outputs feeding directly into downstream warehouse tables.
- Integrated LLM-based document processing using LangChain and RAG into the data platform, enabling AI-driven analytics features that informed client business decisions in a regulated financial environment.
- Collaborated across engineering, product, and analytics teams to define data contracts, build reporting pipelines, and deliver **Metabase dashboards** surfacing business KPIs and pipeline health metrics to stakeholders.
- Led migration of Python and FastAPI prototypes into production **Java Spring Boot microservices**, establishing robust integration between ML inference services and core data platform APIs.

Fleet Logistics Data and Backend Engineer Apr 2023 to Jan 2024

Fleet is a logistics technology company operating fleet management and delivery optimisation platforms across East Africa.

- **Designed and implemented data pipelines** using PostgreSQL to power route optimisation algorithms and operational analytics, contributing to a **30% reduction in delivery time** across the fleet network.
- Integrated the analytics platform with **over five external client systems** via REST APIs, enabling real-time data exchange and supply chain visibility across partner organisations.
- **Automated manual reporting workflows** by building Metabase dashboards and PostgreSQL reporting pipelines, eliminating hours of weekly manual data processing for the operations team.
- Reduced system downtime by **95%** through structured monitoring, root cause analysis, and proactive infrastructure improvements, bringing database query performance and API response times within SLA targets.
- Performed data modelling and schema optimisation on PostgreSQL, improving query performance for high-frequency analytical workloads.

Neural Labs Africa Machine Learning Engineer (Part-time) Apr 2022 to Dec 2023

Neural Labs Africa is a UNICEF Venture Fund-backed MedTech startup whose NeuralSight platform uses computer vision to screen medical images across X-ray, CT, and MRI modalities for over 20 pathologies.

- **Built and containerised FastAPI-based ML inference services** with Docker for scalable deployment on AWS EC2 and SageMaker, enabling real-time model serving for clinical image analysis applications.
- Implemented a complete **MLOps CI/CD pipeline** using Docker, Kubernetes orchestration, and GitHub Actions automation, reducing deployment workload by **80%** and enabling rapid iteration on model versions.
- Used **Apache Spark and HDFS** for distributed pre-processing of large-scale medical imaging datasets, building efficient data loading pipelines for multi-GPU model training workflows.
- Set up **Prometheus and Grafana** for real-time monitoring of model performance and infrastructure health in production environments.
- Migrated the complete ML infrastructure from DigitalOcean to AWS, enabling access to managed compute, storage, and ML services and establishing scalable data and model management practices.

KEY PROJECTS

Automated Financial Document Intelligence Platform Airflow dbt Snowflake Python Tesseract AWS Lambda

- Designed and built an end-to-end data platform for a U.S. bank: ingested raw legal documents, ran OCR extraction, applied dbt transformations, loaded into Snowflake warehouse, and surfaced results through Metabase reporting dashboards. Reduced manual effort by 80% and enabled real-time analytics on previously unstructured financial data.

Logistics Route Optimisation Data Platform PostgreSQL Python REST APIs Metabase

- Designed the data layer powering Fleet's route recommendation engine, including pipeline design, schema modelling, API integrations with over five client systems, and BI dashboards. Delivered a 30% improvement in fleet delivery time and a 95% reduction in system downtime.

Distributed Medical Imaging Data Pipeline Apache Spark HDFS AWS SageMaker Docker Kubernetes

- Built the data engineering backbone for NeuralSight, handling distributed preprocessing of large-scale DICOM medical imaging datasets and feeding production inference systems. MLOps practices introduced cut deployment workload by 80%.

No-Code ML and Data Analysis Application Python Streamlit Scikit-learn Pandas

- Built an interactive web application enabling business users to upload datasets, run exploratory data analysis, and train predictive models without writing code, demonstrating the ability to translate business needs into accessible data tools.

EDUCATION AND CERTIFICATIONS

BSc Computer Science Multimedia University of Kenya 2022

Data Engineer Certification DataCamp Courses include Data Engineering, Supervised Machine Learning, and Python for Data Science

Competitions Zindi Mental Health Prediction for Kenyan Universities MachineHack Electronics Price Prediction
COVID-19 Impact Data Analysis

Industry Recognition Engineering contributor at Neural Labs Africa, a UNICEF Venture Fund alumnus and Villgro Africa AI4Health cohort member, working at the intersection of data engineering and AI for social impact.